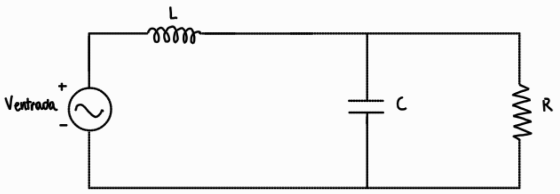
Projeto de filtros passivos - Júlia Emmanuelle Ulbrich

• Filtro passa-baixas (WOOFER) de segunda ordem:



$Z_R = R$

$Z_L = j\omega L$

$Z_C = 1/j\omega C$

$$Z_p = Z_C // R$$

$$Z_p = \frac{Z_C R}{Z_C + R} = \frac{(1/j\omega C) \cdot R}{(1/j\omega C) + R} \cdot \left(\frac{j\omega C}{j\omega C}\right)$$

$$Z_p = \frac{RL}{1 + j\omega CR}$$

divisor de tensão: $$\frac{V_{saída}}{V_{entrada}} = \frac{Z_p}{Z_L + Z_p}$$

$$= \frac{\frac{R}{1 + j\omega CR}}{j\omega L + \frac{R}{1 + j\omega CR}} \cdot \left(\frac{1 + j\omega CR}{1 + j\omega CR}\right)$$

$$= \frac{R}{j\omega L(1 + j\omega CR) + R}$$

$$= \left(\frac{R}{j\omega L + (j\omega)^2 LCR + R}\right) \div R$$

$$= \frac{R/R}{j\omega \frac{L}{R} + (j\omega)^2 \frac{LCR}{R} + \frac{R}{R}}$$

$$H(j\omega) = \frac{1}{1 + j\omega L/R + (j\omega)^2 LC}$$, e dividindo por LC:

função de transferência $$H(j\omega) = \frac{1/LC}{1/LC + \frac{j\omega \frac{L}{R}}{LC} + (j\omega)^2} = \frac{1/LC}{(j\omega)^2 + j\omega/CR + 1/LC}$$ //

Forma padrão: $$H(j\omega) = \frac{\omega_c^2}{(j\omega)^2 + \omega_c/Q\,(j\omega) + \omega_c^2}$$

Comparando termo a termo:

$$\frac{w_c^2}{(jw)^2 + {w_c}/{Q}\,(jw) + w_c^2} = \frac{1/LC}{(jw)^2 + {jw}/{CR} + {1}/{LC}}$$

- $w_c^2 = \dfrac{1}{LC}$ (I)
- $\dfrac{w_c}{Q} = \dfrac{1}{CR_L}$ (II)

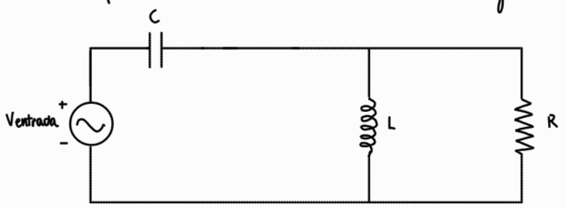
de (II): $C = \dfrac{Q}{w_c R}$ //

em (I): $w_c^2 = \dfrac{1}{L\left(\dfrac{Q}{w_c R}\right)}$

$$w_c^2 = \frac{w_c R}{LQ} \quad \div w_c$$

$$w_c = \frac{R}{LQ} \rightarrow w_c L Q = R \rightarrow L = \frac{R}{w_c Q} \text{ //}$$

• Filtro passa-altas (TWEETER) de segunda ordem:



$Z_R = R$

$Z_L = j\omega L$

$Z_C = 1/j\omega C$

$Z_P = Z_L // R$

$$Z_P = \frac{Z_L \cdot R}{Z_L + R}$$

$$Z_P = \frac{j\omega L R}{j\omega L + R}$$

divisor de tensão: $\dfrac{V_{saída}}{V_{entrada}} = \dfrac{Z_P}{Z_C + Z_P}$

$$= \frac{\dfrac{j\omega L R}{j\omega L + R}}{\dfrac{1}{j\omega C} + \dfrac{j\omega L R}{j\omega L + R}}$$

$$= \frac{\dfrac{j\omega L R}{j\omega L + R}}{\dfrac{1/j\omega C(j\omega L + R) + j\omega L R}{j\omega L + R}}$$

$$= \frac{j\omega L R}{j\omega L + R} \cdot \frac{j\omega L + R}{1/j\omega C(j\omega L + R) + j\omega L R}$$

$$= \frac{j\omega L R}{1/j\omega C(j\omega L + R) + j\omega L R}$$

$$= \frac{j\omega L R}{L/C + R/j\omega C + j\omega L R} \cdot \left(\frac{j\omega C}{j\omega C}\right)$$

$$= \frac{(j\omega)^2 LCR}{R + j\omega L + (j\omega)^2 LCR} \quad \div R$$

$$= \frac{(j\omega)^2 LC}{1 + j\omega \frac{L}{R} + (j\omega)^2 LC} \quad \div LC$$

$$= \frac{(j\omega)^2}{1/LC + j\omega \frac{1}{CR} + (j\omega)^2}$$

função de transferência $H(j\omega) = \dfrac{(j\omega)^2}{(j\omega)^2 + j\omega \frac{1}{CR} + 1/LC}$ //

Forma padrão: $H(j\omega) = \dfrac{(j\omega)^2}{(j\omega)^2 + \omega_c/Q\,(j\omega) + \omega_c^2}$

Comparando termo a termo:

$$\frac{(jw)^2}{(jw)^2 + {}^{wc}/Q\,(jw) + wc^2} = \frac{(jw)^2}{(jw)^2 + jw\frac{1}{CR} + 1/LC}$$

- $\frac{wc}{Q} = \frac{1}{CR}$ (I)

de (I) : $C = \frac{Q}{wcR}$ //

- $wc^2 = \frac{1}{LC}$ (II)

em (II) : $wc^2 = \frac{1}{L\left(\frac{Q}{wcR}\right)}$

$$wc^2 = \frac{wcR}{LQ} \quad \div wc$$

$$wc = \frac{R}{LQ} \rightarrow wcLQ = R \rightarrow L = \frac{R}{wcQ} \text{ //}$$

- Em ambos, $C = \frac{Q}{wcR}$ e $L = \frac{R}{wcQ}$.

$Q = \frac{\sqrt{2}}{2} \approx 0,7071$

$R = 8\,\Omega$

$fc = 2000\,Hz$

$wc = 2\pi fc = 2\pi \cdot 2000 \approx 12566,37$ rad/s

- Então, $C \approx 7,03\,\mu F$

$L \approx 0,9\,mH$